SHRI RAMSWAROOP MEMORIAL UNIVERSITY

Departmental Internal Test- I (2025-26) –Even Semester

Session	**: 2025-26**	**Semester**	**: VI**
Course	**: BCA/Bsc.IT**	**Subject Code**	**: UCS6124**
Branch	**: Normal/AI-DS**	**Subject Name**	**: Machine Learning**
Group	**: ALL**	**Name of Faculty Member**	**: Dr. Archana Maurya, Ms. Aakrati Nigam, Mr. Prem Yadav**

Date	**17/02/2026**	**Duration**	**50 Min.**	**Max. Marks**	**15**	**SET No.**	**I**

University Roll No.															
(To be filled by the Student)															

Note: Please read instructions carefully:

a) *The question paper has 02 sections and it is compulsory to attempt both sections.*

Section A: Short Answer Type Questions **Attempt all the questions.**		**BL**	**CLO**	**Marks (07)**
1.	Define probability theory and statistics. State two differences between them.	BL1	CLO1	01
2.	What is the concept of machine learning? Relate it with a real-world example.	BL1	CLO1	01
3.	Recall the term decision theory and its role in intelligent systems.	BL1	CLO1	01
4.	Why do we need machine learning? Explain the importance of data in machine learning system.	BL1	CLO1	01
5.	Explain how linear algebra forms the mathematical foundation in machine learning.	BL2	CLO1	01
6.	List two differences between machine learning and traditional programming.	BL1	CLO1	01
7.	Name any four popular tools and software frameworks used in machine learning.	BL1	CLO1	01
Section B: Long Answer Type Questions **Attempt all questions.**		**BL**	**CLO**	**Marks (08)**
1.	Explain curve fitting and its significance in machine learning. Discuss underfitting and overfitting.	BL2	CLO1	04
2.	Illustrate the various stages involved in machine learning pipeline in detail with a neat diagram.	BL2	CLO1	04

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Date	**17/02/2026**	**Duration**	**50 Min.**	**Max. Marks**	**15**	**SET No.**	**II**

University Roll No.															
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Note: Please read instructions carefully:

a) *The question paper has 02 sections and it is compulsory to attempt both sections.*

Section A: Short Answer Type Questions **Attempt all the questions.**		**BL**	**CLO**	**Marks (07)**
1.	Find the difference between Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL).	BL1	CLO1	01
2.	Define the reasons for the rapid growth of Machine Learning trends.	BL1	CLO1	01
3.	Tell why mathematics is important for Machine Learning?	BL1	CLO1	01
4.	Name three applications of Machine Learning.	BL1	CLO1	01
5.	Define the concepts of sets in the context of Machine Learning.	BL2	CLO1	01
6.	List the name of four types of machine learning base data analysis	BL1	CLO1	01
7.	Define unsupervised learning.	BL1	CLO1	01
Section B: Long Answer Type Questions **Attempt all questions.**		**BL**	**CLO**	**Marks (08)**
1.	Illustrate how Decision Theory contribute to the development of Machine Learning models.	BL2	CLO1	04
2.	Explain the different types of data analysis performed using machine learning techniques, highlighting their purpose and applications.	BL2	CLO1	04

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Note: Please read instructions carefully:

a) *The question paper has 02 sections and it is compulsory to attempt both sections.*

Section A: Short Answer Type Questions **Attempt all the questions.**		**BL**	**CLO**	**Marks (07)**
1.	Define data in the context of Machine Learning. Name two types of datasets used in Machine Learning.	BL1	CLO1	01
2.	Recall the term probability theory and its role in intelligent systems.	BL1	CLO1	01
3.	List the factors responsible for the rapid growth and increasing adoption of Machine Learning technologies.	BL1	CLO1	01
4.	What is Reinforcement Learning (RL)? List any two applications of Reinforcement Learning.	BL1	CLO1	01
5.	Define the concept of sigmoid function in the context of Machine Learning	BL2	CLO1	01
6.	State the role of linear algebra in Machine Learning.	BL1	CLO1	01
7.	Recall the meaning of over fitting and under fitting.	BL1	CLO1	01
Section B: Long Answer Type Questions **Attempt all questions.**		**BL**	**CLO**	**Marks (08)**
1.	Explain how Information Theory contribute to the development of Machine Learning models.	BL2	CLO1	04
2.	Illustrate the various stages of Machine Learning–based data analysis and explain the significance of each stage.	BL2	CLO1	04